Kingston
School
of Art

LiPo Batteries

Risks and Care

LiPo Battery Risks

🔥 **FIRE & EXPLOSION**

- May catch fire or explode if overcharged, shorted, or punctured.
- Never leave charging batteries unattended.

🎈 **SWELLING**

- Gas buildup from damage or overuse causes swelling—**DO NOT use a swollen battery**.

⚡ **SHORT CIRCUITS**

- Can cause instant overheating and fire.
- Always check for damaged wires or exposed connectors.

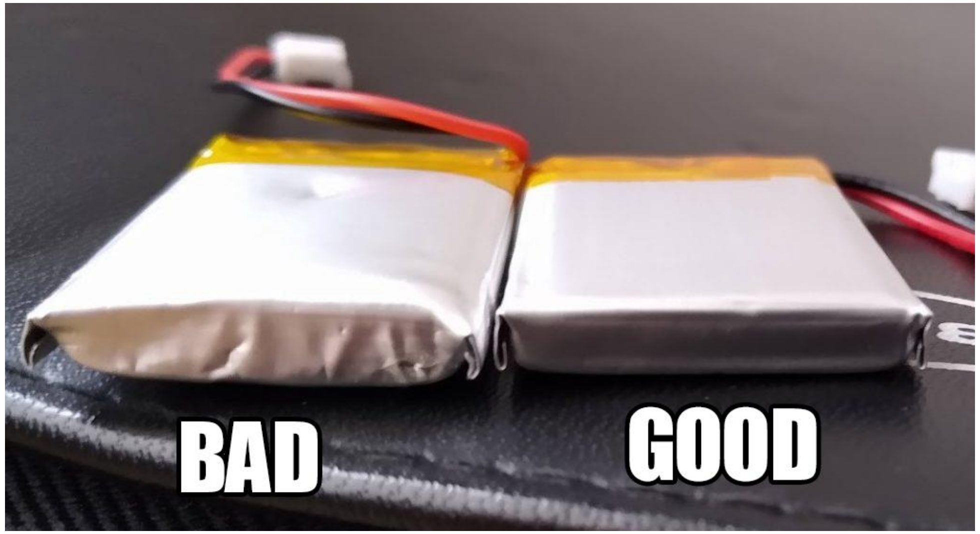
💥 **THERMAL RUNAWAY**

- Once overheated, the battery can self-ignite uncontrollably.
- Move overheating batteries to a **safe, fireproof area immediately**.

🚯 **IMPROPER DISPOSAL**

- Never throw LiPo batteries in the trash.
- Take them to a certified **hazardous waste facility**.

How to Tell if a LiPo Battery is Bad



⚠️ **1. Swelling or Puffing**

- The battery looks bloated or soft.

⚠️ **2. Physical Damage**

- Cracks, punctures, torn wires, or crushed cells.

⚠️ **3. Overheating**

- Gets unusually hot during use or charging.

⚠️ **4. Won't Hold a Charge**

- Charges extremely fast or drains quickly.

⚠️ **5. Low Voltage (Below 3.0V per cell)**

- Measured with a voltmeter or charger.

⚠️ **6. Strange Smell or Smoke**

🔥 *Treat any of the above as an emergency.*

In case of emergency

🔥 IMMEDIATE ACTIONS

- **Stop using the battery immediately.**
  - Disconnect it from the charger, device, or load.
  - Do not try to recharge or use it again until it's fully inspected.
- **Move it to a fire-safe location.**
  - If possible, place the battery outdoors on a **non-flammable surface** (concrete, gravel).
  - **Do not touch it directly**—use heat-resistant gloves or non-conductive tools.
- **Monitor for signs of danger:**
  - **Swelling**, **smoke**, **hissing**, or a **chemical smell** are signs it's about to vent or catch fire.
  - If these occur, **back away immediately** and be ready to use a fire extinguisher (Class D or ABC if D is not available).

🧯 IF IT CATCHES FIRE

- **Do NOT use water.** LiPo fires are **chemical fires**—water can make them worse.
- Use a **fire extinguisher** or **dry sand/dirt** to smother the flames.
- Call emergency services if the fire cannot be safely contained.

? AFTER IT COOLS DOWN

- If no fire or explosion occurred:
- Place the battery in a **LiPo-safe bag or fireproof container.**
- **Do not reuse the battery.** It's no longer safe.
- Take it to a **hazardous waste disposal center** or an electronics store that accepts batteries.

🚫 WHAT NOT TO DO

- Don't puncture, compress, or shake the battery.
- Don't put it in the trash.
- Don't try to revive it or use it again.

How to take care of a LiPo battery

1. Charge Safely

- Use a **LiPo-compatible charger** with the correct voltage and current settings.
- Do NOT use the battery near any source of heat.

2. Store Properly

- Store batteries in a **cool, dry place**, away from direct sunlight and heat sources.
- **Avoid storing in extreme temperatures**; both high and low temperatures can degrade battery performance.
- **Do not immerse in liquids** or expose to moisture.
- Always store your batteries in an enclosure free of sharp objects. When installing a battery in your project, take care to keep it away from parts of your project that could pinch, poke, or strain the battery.

3. Handle / Transport with Care

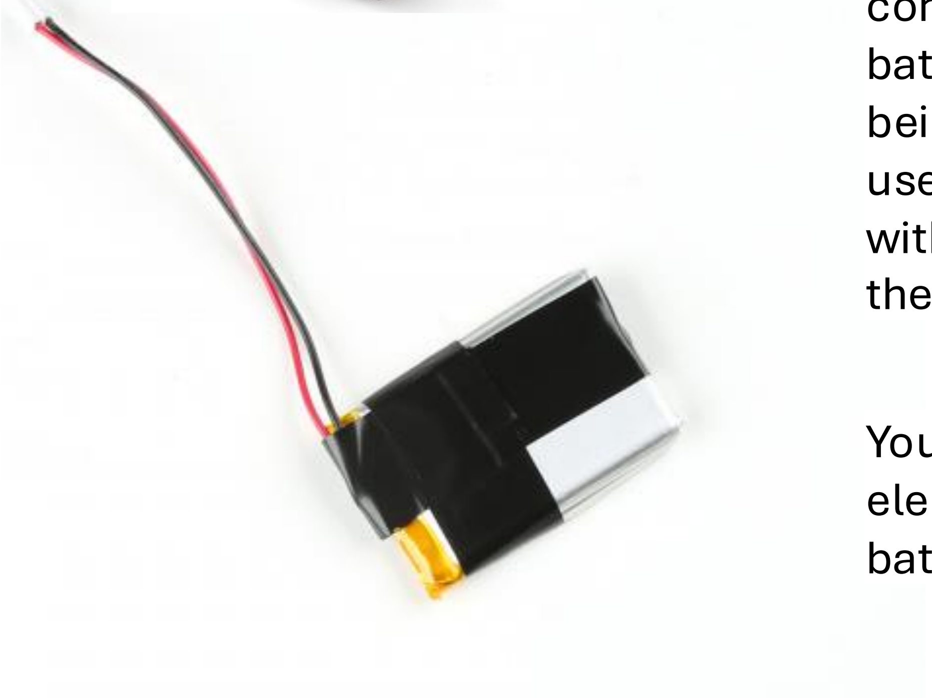
- Avoid dropping or puncturing the battery.
- Do not transport or store a LiPo battery with metal objects, such as hairpins, necklaces, or any other conductive object or material.

4. Inspect battery before each use

- Check for heat, puffiness or any other visible damage.

5. Dispose Responsibly

- Recycle at an **approved hazardous waste center**.
- *Never throw LiPos in the trash.*



Recommended

One of the down sides to using LiPo batteries is their fragile power connections between the wires and protection circuit. These type of batteries are manufactured for a permanent install in devices, and not being removed often as can sometimes happen with projects. For users prototyping or frequently removing the battery from boards without ON/OFF switches, it can be easy to accidentally pull or break the power wires from the terminals.

You can provide strain relief to the wires by securing them with electrical tape - this will help with strain on the connection to the battery.

Further Reading

LiPo Battery Care by Sparkfun:

- https://learn.sparkfun.com/tutorials/single-cell-lipo-battery-care

LiPo Battery Care by Adafruit:

- https://learn.adafruit.com/li-ion-and-lipoly-batteries/overview